a. Mitosis – Stage : Metaphase

Diagnostic features:

- The spindle fibres attached to the kinetochore region of centromere of chromosomes
- Chromosomes are arranged at the equator region of the cell (metaphase plate)
- Chromosomes are distinctly visible in this stage.

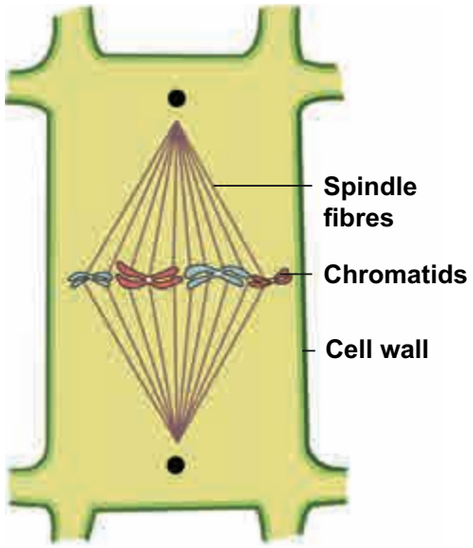


Figure 4a: Metaphase

b. Mitosis – Stage : Anaphase

Diagnostic features:

- Each chromosome splits and two daughter chromatids begin to move towards opposite poles.
- Shortening of spindle fibre and longitudinal splitting of centromere creates a pull which divide the chromosomes.

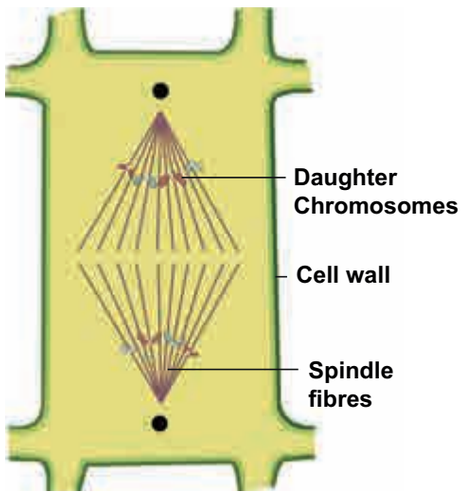


Figure 4b: Anaphase

Exercise: 5

Plant anatomical structures (Dicot-Root, Stem and Leaf, Monocot- Root, Stem and Leaf)

Aim: To study and identify the T.S of dicot root, dicot stem, dicot leaf, monocot root, monocot stem and monocot leaf.

Principle: A group of tissues performing a similar function, irrespective of its position in the plant body, is called a tissue system. The three types of tissue system in plants are: Epidermal tissue system, Ground tissue system and Vascular tissue system. In different parts of the plants, the various tissues are distributed in characteristics patterns. This is

11th Botany_EM_Chapter 15.indd 346 11/01/2023 15:00:42